Inequalities — problem 7.85

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Let $x_1, \ldots, x_n$ be positive with $x_1 \cdot x_2 \cdots x_n = 1$. Prove that

$$x_1^{n-1} + x_2^{n-1} + \cdots + x_n^{n-1} \geq \frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}.$$